

Do **not** write solutions on this page.

11. [Maximum mark: 22]

Consider the complex number $w = \frac{4i}{1+i\sqrt{3}}$.

(a) Show that $w = \sqrt{3} + i$. [2]

(b) Find

(i) $|w|$;

(ii) $\arg w$. [3]

(c) Use De Moivre's theorem to find the two square roots of w . Give your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]

The complex number w can be expressed in the form $(a + bi)^2$, where $a, b \in \mathbb{R}^+$.

(d) (i) Show that $a^2 = \frac{\sqrt{3}}{2} + 1$.

(ii) Find the value of b^2 . [8]

(e) Hence, express $\tan \frac{\pi}{12}$ in the form $p + q\sqrt{3}$, where $p, q \in \mathbb{Z}$. [5]

